Encode London, 25 October 2026 · github.com/anushaisawesome/unseen-steps · unseen-steps-studio.vercel.app

Test report: ready for real listeners

What we tested before submitting, how, and what it showed, including the limits of each test.

0
axe WCAG 2.2 AA violations, 3 pages × light/dark × 375/1440 px

14/18
venue describe calls returned a line in the screen-reader test (the rest chose silence)

311/311
narration clips say exactly their caption

0.81 s
median live latency, 33 lines across two sessions

1. Accessibility of the site

Method: Playwright (Chromium) with axe-core 4.10.2, rule sets WCAG 2.0/2.1/2.2 A and AA plus best practice, run on the home page, the venue page and the 18 styles page, each in light and dark colour schemes at 375 px and 1440 px wide, with reduced motion on.

Check	Result
axe violations	0 on every page, scheme and width (12 runs)
Sideways scrolling at 375 px	None on any page
Keyboard order (venue page)	Skip link, logo, Live at the venue, 18 styles, How it works, Ballet solo, Back to home, then the settings in reading order and Start listening
Touch targets	No focusable control under 24 px (WCAG 2.2 target size)
Headings	One h1 per page with a clean h2/h3 outline
Captions	Narration shown as resizable captions, a transcript and a WebVTT download; an opt-in live region sends lines to a screen reader

2. Venue page with a simulated screen reader

Method (`tests/venue_screenreader_test.py`): the venue page served locally with its two live functions (`scripts/dev_server.py`), an iPhone 14 viewport with touch, a fake camera playing Andrii's own rehearsal footage, and keyboard-only navigation: Tab through the page, turn on "Also send lines to my screen reader", press Enter on Start listening, stop after about 45 s. An overlay lists each focus change (role, name, state) and each live-region announcement, computed from the accessibility tree.

Results: axe 0 WCAG 2.2 AA violations on the page; every control announced with a role and a name (for example "radio button, Detailed: every new line, selected"; "slider, Narration volume, 100%"); status messages announced (asking for the camera, starting the pose model, listening); 18 describe calls, 14 returned a line and 4 chose silence; median round trip 2.3 s. Sample announcements: "Holding a still frame for an imaginary partner." "Steps backward, maintaining the frame."

> **Limit:** the screen reader is simulated from the accessibility tree; it is not VoiceOver or TalkBack, and the recording has no sound. A recording on an iPhone with VoiceOver is the real-device check. The venue page's lighter model repeated lines in this run.

3. Narration clips against their captions

Every clip on the 18 styles page (311) was transcribed by Gemini and compared with its caption, with up to three checks per clip. **311 of 311 match**; no transcript mentions director's notes, tone or delivery; no line starts too late

to be heard; three last lines run 0.4 to 1.1 s past the clip's end and are allowed to finish. The home page's ballet narration was re-voiced and transcribed: all five lines match word for word.

4. Live sessions

Session	Lines	First sound after the prompt	Checker flags
6	20	median 0.79 s (0.62 to 0.99)	3, two of them false flags since fixed ("the right moment", an unmeasured rotation)
7	13	median 0.81 s (0.61 to 1.24)	1: "Natural spin turn, rotating rightwards", flagged "it says right, but the measured travel was left"

5. MongoDB Atlas features

- **Atlas Search** (fuzzy `$search` on song names): "andri walts" finds "Andrii's waltz (rehearsal soundtrack)", 86.1 bpm, 3/4; "tango" finds nothing; the live program loads the stored map with its 29 gaps.
- **Hybrid Search** (Vector Search and Atlas Search fused by reciprocal rank, vetted terms only): "rotates to the right as a couple, a natural turn" ranks natural turn first in both; "steps forward, side, then closes the feet together" ranks closed change first.

How to rerun

- `python scripts/dev_server.py` then `python tests/venue_screenreader_test.py`: the venue page test and its video.
- `python scripts/regress.py`: a dancer's must-haves per demo, judged from the timed lines.
- Live sessions: `python scripts/live.py --show "Demo: Andrii, waltz" --song "andri walts" --camera 0 --beats-per-bar 3 --style waltz --record night.mp4` writes the log used above.